

Automated KYC for Vibe House

1. Summary

Cloud-based KYC solution - requirements designed to automate the pre-checkin and check-in process.

2. Use Case 1: Pre-Checkin (Remote Guest Onboarding)

2.1 Overview

The guest completes this flow on their own smartphone via a link sent post-booking.

2.2 Detailed Action Sequence

Step	User Action (Click/Type)	System Response / Backend Logic
1. Landing	Click the URL in SMS/WhatsApp.	Authenticate <code>booking_id</code> ; Load Property Branding & Welcome Screen.
2. Auth	Type last 4 digits of registered mobile number.	Validate against DB record; Grant session access.
3. Type Selection	Click "Indian Citizen" or "International Guest".	Set <code>nationality_type</code> variable; Toggle relevant legal fields.
4. ID Selection	Click dropdown: Select Aadhaar, Passport, or DL.	Update UI labels for the upload section.
5. Front Upload	Click "Upload Front" button.	Open mobile camera/gallery; Upload to DB Storage.
6. Back Upload	Click "Upload Back" button (if Indian ID).	Open camera/gallery; Upload to DB Storage.

7. OCR Trigger	Automatically when both uploads are successful	Call Vision API (OCR) on uploaded images.
8. Data Review	Type/Edit: Name, DOB, ID Number, Address.	Pre-fill fields with OCR data; Highlight low-confidence characters. Retry if OCR Fails
9. Travel Info	Type: "Coming From", "Going To", "Purpose".	Fields mandatory for Form C/Police records.
10. Finish	Click "Submit & Save".	Update status to PRE_VERIFIED ; Show "See you at the property" screen and send confirmation notification (Email or WA).

3. Use Case 2: Onsite Check-in (Tablet-Based Kiosk)

3.1 Overview

The guest interacts with a mounted tablet at the property entrance.

3.2 Detailed Action Sequence

Step	User Action (Click/Type)	System Response / Backend Logic
1. Start	Click "Start Check-in" button on Tablet.	Reset session; Wake up camera hardware.
2. Lookup	Type Mobile Number or Scan Booking QR.	Query DB; Pull PRE_VERIFIED data & remote ID images.
3. Face Match	Click "Take Selfie".	Liveness Check: Prompt user to "Blink". Capture frame.

4. AI Compare	(Wait for processing).	Vision API: Compare live selfie vs. photo on Remote ID.
5. ID Verification	Click "Scan Physical ID".	Open high-res back camera. Overlay "Place ID here" guide.
6. Capture	Hold ID steady (Auto-capture or Click).	Capture "Onsite Scan". Compare text/layout with "Remote Upload".
7. G-Card Review	Scroll through the generated PDF G-Card.	Populate Template with all data (Name, ID, Address, Travel).
8. Signing	Use finger to sign in the "Guest Signature" box.	Convert canvas drawing to PNG; Embed into final G-Card PDF.
9. Final Submit	Click "Check-in Now".	Update DB status to <code>CHECKED_IN</code> ; Log timestamp.
10. Access	Read Room No. and Passcode from screen.	Smart Lock API: Generate unique code for <code>booking_id</code> .

4. Technical Specifications & Error Handling

4.1 Match Logic (Vision AI)

- **Face Match:** Required Score > 0.85. If < 0.85, allow 1 retry, then flag for manual override.
- **Document Comparison:** System compares the OCR text of the Onsite ID Scan with the Remote ID Scan. If the ID Number or Name differs, the flow is blocked.

4.3 Edge Case Handlers

- **Input Blur:** If Onsite ID Scan is blurry, the "Submit" button remains disabled until a sharp frame is detected.
- **Smart Lock Offline:** If the Lock API fails, display "Staff notified. Please collect a physical key" and send a P1 alert to the Manager.

5. Compliance Checklist

- **Consent:** Guest must click a mandatory checkbox: *"I consent to the digital processing of my ID for check-in purposes"* during the Pre-checkin phase.

6. Hardware Specifications

To ensure smooth AI processing (Face Match) and clear document capture, the following minimum specifications are mandated for property rollout:

- **Device:** 11-inch or 12.9-inch Tablet (iPad Air/Pro or Samsung Galaxy Tab S9).
- **Camera:** Minimum 12MP Front Camera with Wide-Angle support (for selfies) and 12MP Rear Camera (for document scanning).
- **Processor:** Apple M-series or Snapdragon 8 Gen 2 (required for local liveness processing and fast OCR).
- **Mounting:** Desktop or Floor-stand enclosure with integrated charging and a physical lock to prevent port tampering.
- **Connectivity:** Wi-Fi 6 + 4G/5G Failover (SIM card slot required for properties with unstable Wi-Fi).

7. Kiosk Environment & Deployment

The tablet must run in a "Locked Kiosk Mode" to prevent guests from exiting the app or changing system settings.

7.1 Managed Mode (MDM)

- **Tool:** Use an MDM (Mobile Device Management) solution like **Microsoft Intune**, **Jamf**, or **Google Workspace Endpoint Management**.
- **Configuration:**
 - **Single App Mode:** The tablet is locked to the Sarathi Web App URL.
 - **Disable Notifications:** All system alerts, OS updates, and pop-ups are suppressed.
 - **Power Management:** Set to "Never Sleep" while connected to power; auto-dim during property quiet hours (e.g., 2 AM - 6 AM).

7.2 Progressive Web App (PWA) Architecture

The application will be deployed as a PWA to ensure:

- **Full-screen UI:** Removes browser address bars and navigation buttons.
- **Service Workers:** Basic assets (CSS, JS, Logos) are cached locally so the UI loads instantly even on slow connections.
- **Hardware Access:** Explicit permissions for Camera and GPS are granted at the MDM level so guests aren't prompted.

7.3 On-Device Processing vs. Cloud

- **Liveness Detection:** Performed on-device using a lightweight TensorFlow.js model to detect eye-blinks or head movement. This reduces latency and bandwidth.

- **Face Match & OCR:** The high-res frames are sent to **Vision API** via Firebase Functions.
- **Image Comparison:** The "Onsite Scan" is compared against the "Remote Upload" using a Structural Similarity Index (SSIM) algorithm to detect if the physical document is a photocopy or the original.

7.4. Security & Compliance

- **Session Purge:** Upon "Check-in Complete" or "Session Timeout," all local blobs/caches of guest photos are wiped from the tablet's memory.
- **Encrypted Transit:** All data sent via HTTPS (TLS 1.3).
- **Physical Security:** The stand must hide the "Home" button and block USB ports to prevent unauthorized data extraction.

8. Operational Maintenance (Ops Team)

8.1 Monitoring by Property manager:

- Real-time status of all tablets (Online/Offline, Battery Level).
- **Success Rate:** Percentage of guests completing check-in without staff help.